

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 17307

LM SUPERCONDUCTIVITY

SUMMER EXAMINATION 2024

Time Allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.
Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks.
Answer ***two*** questions from this Section. If you answer more than two questions,
credit will only be given for the best two answers.

The approximate allocation of marks to each part
of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text.
Calculators with the ability to store text should have their memories deleted prior to
the start of the examination.

A formula sheet and a table of physical constants and units that may be required will
be found at the end of this question paper.

SECTION 1

Answer **all four** questions from this Section.

1. Explain briefly what is meant by the Meissner effect, and show that it may be understood using the London equations

$$\begin{aligned}\nabla \times \mathbf{J}_s &= -\frac{n_s e^2}{m} \mathbf{B}, \\ \frac{\partial \mathbf{J}_s}{\partial t} &= \frac{n_s e^2}{m} \mathbf{E},\end{aligned}$$

where $\mathbf{J}_s$ is the supercurrent density.

[10]

2. Use the expression for the supercurrent density

$$\mathbf{J}_s = \frac{n_s^* e^*}{m^*} (\hbar \nabla \phi - e^* \mathbf{A}),$$

to show that the fluxoid threading a hole in a multiply-connected superconductor,

$$\Phi' = \int_S \mathbf{B} \cdot d\mathbf{S} + \frac{m^*}{n_s^* e^{*2}} \oint_C \mathbf{J}_s \cdot d\mathbf{r},$$

is quantised in units of h/e^* .

[10]

3. Draw the magnetisation curves as a function of external applied magnetic field, $M(H)$, for type-I and type-II superconductors. Explain what happens physically at significant points on the two curves.

[10]

ANY CALCULATOR

4. The order parameter, $\psi(x)$, of a superconducting wire satisfies the Ginzburg-Landau equation

$$-\frac{\hbar^2}{2m^*} \frac{d^2\psi}{dx^2} + a\psi + b|\psi|^2\psi = 0.$$

By considering a current-carrying solution, $\psi(x) = \psi_0 e^{ikx}$, derive a formula for the critical current density, J_c , which is the maximum value of the supercurrent density, $J_s = n_s^* e^* v_s$.

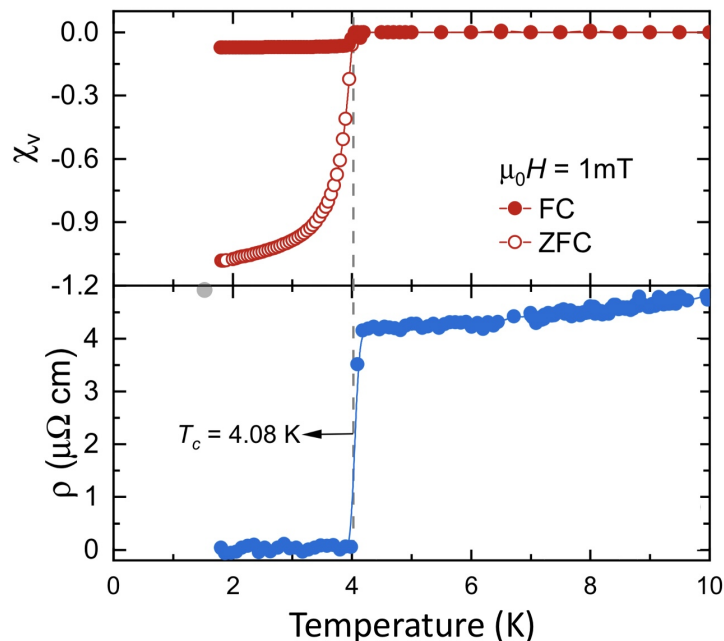
[10]

SECTION 2

Answer **two** questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.

5. Suppose that we have performed experiments on an ellipsoidal superconductor to measure magnetic susceptibility χ_v (assuming linearity), magnetisation M and electrical resistivity ρ in a range of temperature (T) and a magnetic field (H), where the field is applied along a principal axis.

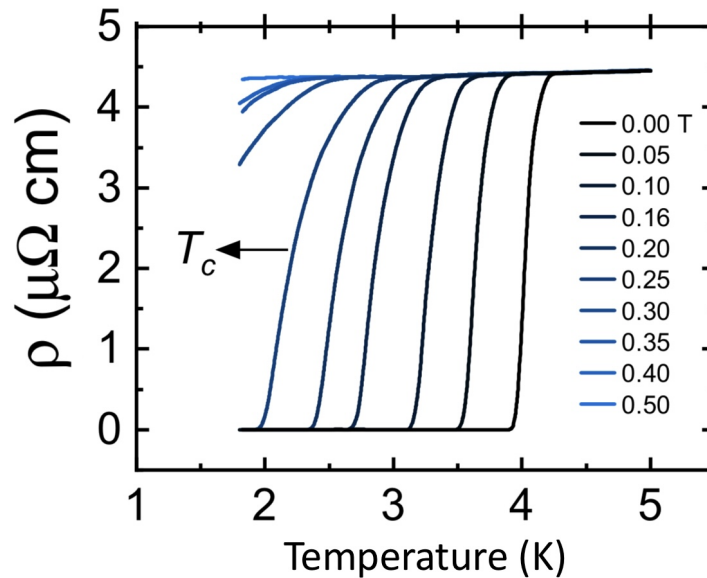
- (a) The graphs below plot the susceptibility (upper panel) and resistivity (lower panel) data as a function of temperature. ZFC and FC refer to zero-field-cooling and field-cooling conditions, respectively while the measurements were taken while warming up.



Briefly explain i) why there is a difference in the ZFC and FC susceptibility ii) why the ZFC susceptibility at the lowest measured temperature is less than -1 for a perfect diamagnet and iii) why the resistivity drops more abruptly than the susceptibility on cooling across T_c .

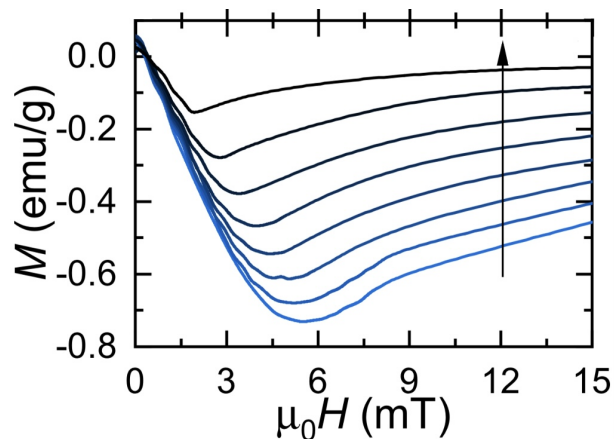
[9]

- (b) The graph below plots the resistivity as a function of temperature in different field strengths.



Explain the implication of the observation that the temperature where the resistivity curve $\rho(T)$ reaches zero decreases as the field strength increases, and also that the curve does not reach zero in 0.30 T or higher field strength, becoming essentially flat in 0.50 T. [5]

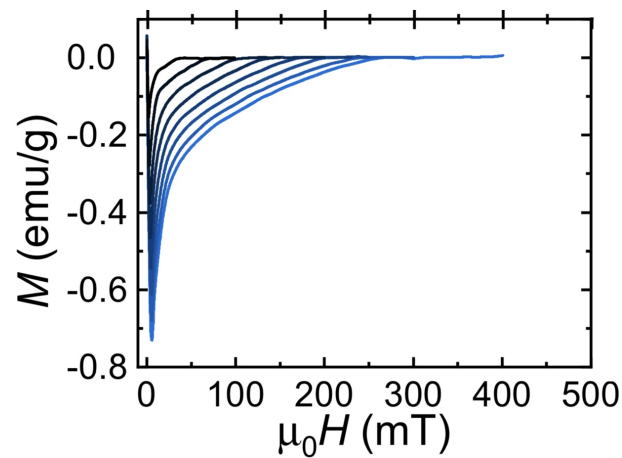
- (c) The graph below plots magnetisation as a function of an applied magnetic field at different temperatures between 2 and 3.75 K that is varied in equal steps, where the arrow indicates the order of measurements at different temperatures; the lightest coloured curve being measured first at one temperature while the darkest one measured last at the other temperature.



Explain which of the curves, the lightest or the darkest, should correspond to the lowest measured temperature and why. Estimate the first critical field strength at zero temperature $\mu_0 H_{c1}(0)$. [8]

ANY CALCULATOR

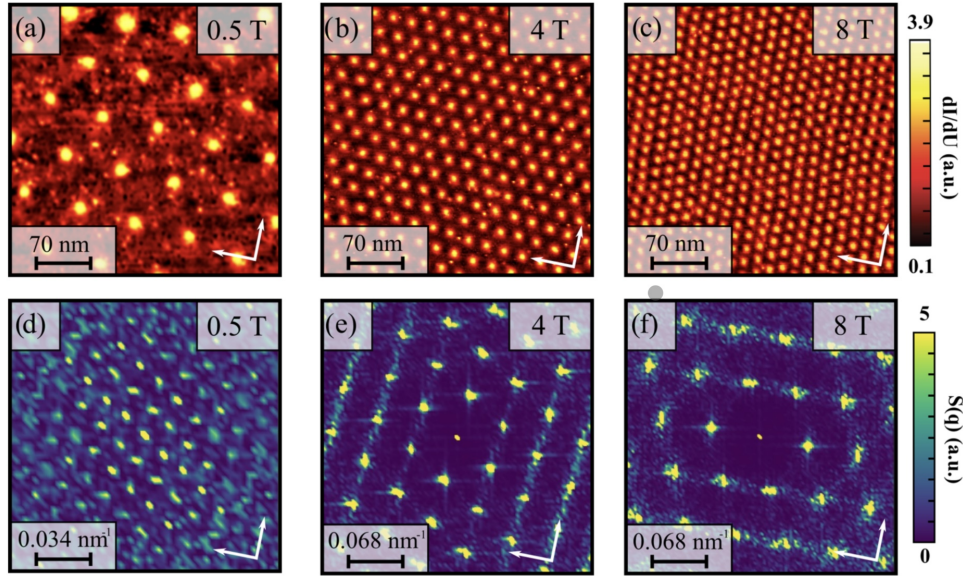
- (d) The graph below plots magnetisation as a function of an applied magnetic field, over a wider range than in the previous question (c), at different temperatures between 2 and 3.75 K varying in equal steps.



Estimate the zero-temperature second critical field strength $\mu_0 H_{c2}(0)$ and then evaluate the zero-temperature coherence length $\xi(0)$ within the Ginzburg-Landau framework. **[8]**

6. Suppose that we have performed experiments on a type-II superconductor to obtain information about its flux lattice, coherence length ξ and penetration depth λ .

(a) The images below show differential tunneling conductance (dI/dU) map in real space on the surface of the superconductor (upper panel) and its Fourier transform $S(q)$ (lower panel) in different applied field strengths of 0.5, 4 and 8 T. (Note that the images have not necessarily been taken at the same position in space and a.u. represents arbitrary unit)



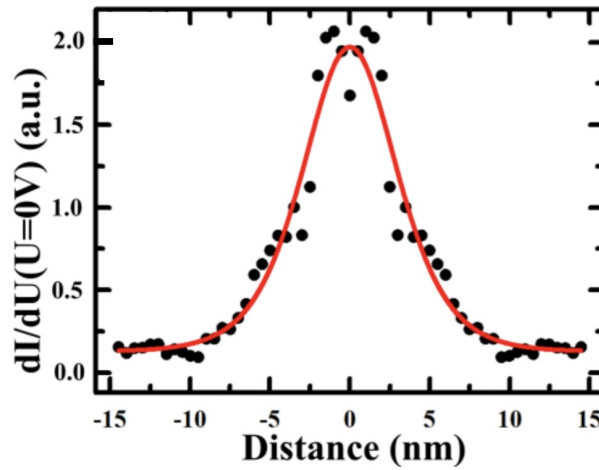
Explain whether the lattice type (e.g., triangular or square) has changed as a function of the applied field strength or not and why. Also explain why the density of the flux lines has increased on increasing the applied field strength.

[8]

- (b) The superconducting order parameter across a vortex can be written as

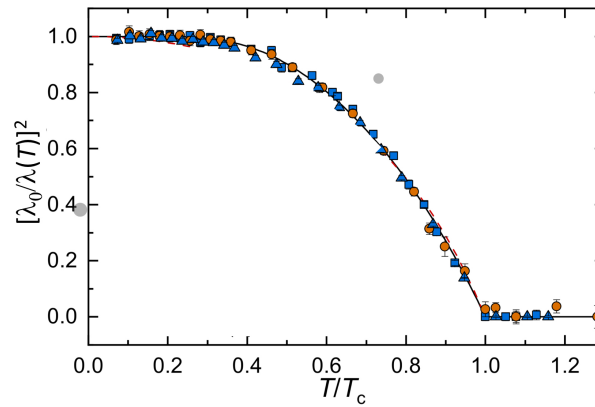
$$|\psi(r)| = |\psi_\infty| \tanh\left(\frac{r}{\xi_{GL}}\right),$$

where r is the distance from the centre of a vortex. Assuming that the total density of electrons is a sum of the normal and superconducting electrons, express $dI/dU(U = 0 \text{ V})$ in terms of the coherence length, ξ_{GL} . Using the obtained expression, estimate ξ from the $dI/dU(U = 0 \text{ V})$ from the graph below.



[12]

- (c) The graph below plots the penetration depth λ with temperature T , where λ_0 is the penetration depth at zero temperature and T_c is the critical temperature for the superconducting transition.



What does the almost temperature independent behaviour of $[\lambda_0/\lambda(T)]^2$ below $T/T_c \simeq 0.3$ imply? Explain in terms of other relevant physical quantities.

[10]

7. In Ginzburg-Landau (GL) theory, the difference in the free energy density between the superconducting (f_s) and normal (f_n) states is given by

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi|^2,$$

where the order parameter $\psi = |\psi|e^{i\phi}$. The supercurrent density can be expressed as

$$\mathbf{J}_s = -\frac{e^*|\psi|^2}{m^*} (\hbar\nabla\phi + e^*\mathbf{A}) = -e^*|\psi|^2\mathbf{v}_s,$$

where $\mathbf{v}_s$ is velocity of the superconducting charge carriers.

Consider a type-I superconductor slab whose thickness d is small compared to the coherence length such that spatial variation of the amplitude $|\psi|$ can be neglected.

- (a) By minimising the difference in the free energy density, obtain the following expression for the supercurrent density in the absence of applied magnetic field,

$$\mathbf{J}_s = -e^*|\psi_0|^2 \left[1 - \left(\frac{\xi m^* v_s}{\hbar} \right)^2 \right] \mathbf{v}_s,$$

where the Ginzburg-Landau coherence length $\xi = \hbar/\sqrt{2m^*|a|}$ and $|\psi_0|^2 = -a/b$. Also show via $\partial J_s/\partial v_s = 0$ that the maximum supercurrent density is obtained as

$$J_{s,\max} = \frac{2e^*\hbar n_s^*}{3\sqrt{3}m^*\xi},$$

and express the corresponding momentum (i.e., when $J_s = J_{s,\max}$) in terms of $\hbar$ and ξ .

[10]

- (b) Let us identify the above maximum supercurrent density $J_{s,\max}$ with the critical current density J_c . In London theory, a separate estimate for J_c is found when the kinetic energy becomes equal to the condensation energy (the magnetic energy of the critical field). By equating the kinetic and magnetic energy densities, obtain

$$J_c = H_c/\lambda,$$

where the penetration depth λ is given by

$$\lambda = \left(\frac{m^*}{\mu_0 n_s^* e^{*2}} \right)^{1/2}.$$

ANY CALCULATOR

Also compare quantitatively the two J_c values from the GL and London theories to state which one is larger, by how much, and then explain the reason.

(Hint: you may use the relation $-a^2/(2b) = -\mu_0 H_c^2/2$.)

[10]

- (c) Suppose that a magnetic field H is applied parallel to the surface of the slab that is defined by the plane boundaries $x = \pm d/2$. By minimising the free energy per unit area (i.e., free energy density integrated over the thickness of the slab) with respect to $|\psi|^2$ while using the gauge $\mathbf{A} = (0, \mu_0 H x, 0)$, show that the critical field H'_c has increased from the bulk value H_c as

$$H'_c = 2\sqrt{6}H_c \left(\frac{\lambda}{d} \right).$$

(Hint: one would expect $|\psi|^2 \rightarrow 0$ as H approaches H'_c).

[10]

Formula Sheet

Vector identity: $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$.

Maxwell's equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}, \quad (4)$$

Some expressions from the Ginzburg-Landau (GL) theory:

$$\lambda = \sqrt{m^* b / 4 \mu_0 e^2 |a|},$$

$$\xi = \sqrt{\hbar^2 / 2 m^* |a|},$$

$$B_{c2} = \frac{\Phi_0}{2\pi\xi^2} = \sqrt{2}\kappa B_c,$$

where a and b are the GL parameters, Φ is the flux quantum and $\kappa \equiv \lambda/\xi$.

The GL free energy density difference is expressed as

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi|^2,$$

where f_s and f_n are the free energy density in the superconducting and normal states, respectively. Also note that $n^* = |\psi|^2$.

Some predictions from the BCS theory:

$$\xi_0 = \frac{\hbar v_F}{3.52 k_B T_c},$$

$$\Delta_0 = 1.76 k_B T_c,$$

$$k_B T_c = 1.14 k_B \Theta_D \exp[-1/D(E_F)V],$$

where $D(E_F)$ is the density of states at the Fermi level and V the effective electron-electron attraction via phonons.

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.